Exact-self-intersection orbit counts: a peripheral-block construction

Research note prepared for Joe

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Abstract

For the once-punctured torus and each fixed m-punctured sphere, $m \geq 3$, we give a proof that the number of mapping-class-group orbits with *exactly* K minimal self-intersections is $\exp(\Theta(\sqrt{K}))$. The lower bound is a deduction from a peripheral-block padding lemma, the Diop–Gaye intersection formula, and the Chas–Lalley central limit theorem. Its leading exponential coefficient is at least $3 \log 3$. The upper bound is the punctured-surface extension of Aougab–Souto. The bounds do not identify the exact leading coefficient, prove that its limit exists, or give an asymptotic equivalent. No priority claim is made for the deduction.

1 Conventions and result

Let $S_{g,b}$ denote the interior of a compact orientable surface of genus g with b boundary components, and let $\mathrm{si}(\gamma)$ be the minimum number of transverse double points in general position in its free homotopy class. Set

$$\mathcal{O}(K,S) = \#\{\text{mapping-class orbits with } \mathrm{si}(\gamma) \leq K\},$$
$$\widetilde{\mathcal{O}}(K,S) = \#\{\text{mapping-class orbits with } \mathrm{si}(\gamma) = K\}.$$

The objective on page 5 of the supplied notes distinguishes these two quantities. Throughout, log is natural. Here *primitive* means not a proper power; it does not mean membership in a free basis. The lower bound below is valid even after quotienting by orientation-reversing homeomorphisms and reversal of the curve. Using only orientation-preserving mapping classes, orienting the curves, or labeling the punctures changes fixed-surface counts by bounded factors. Boundary twists act trivially on closed free homotopy classes; therefore replacing punctures by boundary components fixed pointwise also changes the relevant counts by at most a fixed puncture-labeling factor.

Theorem 1. *Let $S = S_{1,1}$ or $S = S_{0,m}$ for a fixed $m \geq 3$. There are constants $c > 0$ and K_0 such that, for every integer $K \geq K_0$,*

$$\widetilde{\mathcal{O}}(K,S) \geq \frac{c}{\sqrt{K}}\, 3^{\,3\sqrt{K} - 9K^{1/4}}. \tag{1}$$

For every $\varepsilon > 0$, for all sufficiently large K,

$$\widetilde{\mathcal{O}}(K,S) \leq \mathcal{O}(K,S) \leq \exp\Big(\big(4\sqrt{2|\chi(S)|} + \varepsilon\big)\sqrt{K} \Big). \tag{2}$$

Consequently

$$\log \widetilde{\mathcal{O}}(K,S) = \Theta_S(\sqrt{K}), \qquad \log \mathcal{O}(K,S) = \Theta_S(\sqrt{K}),$$

and

$$3\log 3 \le \liminf_{K\to\infty} \frac{\log \widetilde{\mathcal{O}}(K,S)}{\sqrt{K}} \le \limsup_{K\to\infty} \frac{\log \widetilde{\mathcal{O}}(K,S)}{\sqrt{K}} \le 4\sqrt{2|\chi(S)|}.$$

The lower-bound curves are primitive and nonperipheral.

For $S_{0,3}$ and $S_{1,1}$ the two displayed leading constants are $3\log 3 \simeq 3.29584$ and $4\sqrt{2} \simeq 5.65685$. Thus the result is sharp in the power $\sqrt{K}$ of the exponent, not in its leading coefficient. It is not an assertion that $\widetilde{\mathcal{O}}(K,S) = \Theta(e^{c_S\sqrt{K}})$ for any specified c_S.

2 Two published inputs for the lower bound

Let P be a pair of pants. Choose the usual geometric free basis a, b consisting of peripheral loops, and write $A = a^{-1}$, $B = b^{-1}$. One can use the cyclic Schottky alphabet a, B, b, A of [3]; changing orientations of the basis letters is harmless. A *block* is a maximal run of one letter in a cyclically reduced word.

Intersection formula. For a primitive cyclically reduced word involving both generators, write

$$w = s_1^{p_1} t_1^{q_1} \cdots s_r^{p_r} t_r^{q_r}, \quad s_i \in \{a, A\}, \quad t_i \in \{b, B\}, \quad p_i, q_i > 0.$$

Let $N = |w|$. Diop–Gaye, Theorem 3.12 of the preprint [3], gives

$$\mathrm{si}(w) = H(w) + rN - 2r^2 - \sum_{1\le i<j\le r} \big(|p_i - p_j| + |q_i - q_j|\big). \tag{3}$$

The integer $H(w)$ is determined by intersections of lifts based at block boundaries and certain equalities of signed blocks. Appendix A specifies it completely. Its dependence on endpoints can be decided from first-differing-letter comparisons in the planar free-group boundary.

Random-word theorem. Let $\mathcal{F}_N$ be the set of oriented cyclically reduced conjugacy classes of length N in this geometric basis. For a uniform random $W_N \in \mathcal{F}_N$, the main theorem of Chas–Lalley [2] gives, on a pair of pants,

$$\frac{\mathrm{si}(W_N) - N^2/9}{N^{3/2}} \implies \mathcal{N}(0, 1/81). \tag{4}$$

Here their general constants are $\kappa = \chi/[3(2\chi-1)]$ and $\sigma^2 = 2\chi(2\chi^2 - 2\chi + 1)/[45(2\chi-1)^2(\chi-1)]$; substitution of $\chi(P) = -1$ yields the constants above. Only a fixed upper-tail consequence of this theorem will be used; no local limit theorem at an exact intersection number is assumed.

3 The padding lemma

Lemma 2 (Unique-longest-block padding)**.** *Suppose the cyclic word $w_M = a^M u$ is reduced, u begins and ends with b or B, and every a- or A-block of u has length strictly smaller than M. Then for every integer $d \ge 0$,*

$$\mathrm{si}(a^{M+d}u) = \mathrm{si}(a^M u) + d. \tag{5}$$

All these words are primitive.

Proof. There is exactly one longest block involving a or A. A proper power would repeat every cyclic block, including that block. Hence w_M and every padded word are primitive.

We first show that $H(w_M)$ is constant as M increases while the other blocks remain fixed. The distinguished lifts used in H start at block boundaries. Their forward endpoints are infinite periodic words read forward from these boundaries; their backward endpoints are the corresponding inverse periodic words. The cyclic order of endpoints is determined by the first letters at which their rays differ, with the local order prescribed by the last common letter.

Compare any two distinct such rays. Their matching initial blocks have equal signed lengths. If their first mismatch does not involve the distinguished block, neither the common last letter nor the two differing letters changes when M grows. If the first mismatch does involve that block, the competing block has strictly smaller length. The mismatch therefore occurs when the shorter block ends, regardless of the value of M: one ray continues with a (or A when reading backward), while the other takes the same next letter as before. The local ordering is again unchanged. Two distinct forward rays cannot encounter the unique distinguished block simultaneously after a matching block prefix: that would identify their cyclic block starting positions. The same holds for two backward rays. A forward distinguished block is a^M, whereas the backward distinguished block is A^M, so opposite-direction rays cannot match those two distinguished blocks to each other. Equal rays, if any are listed redundantly, remain equal.

Thus all endpoint cyclic orders, and hence all crossing indicators in H, are unchanged. The signed-block equality conditions in the definition of H are unchanged as well: the distinguished exponent remains strictly larger than every competing a-exponent. Therefore $H(w_{M+d}) = H(w_M)$.

Now apply (3), with $p_1 = M$. The term rN increases by rd. Exactly $r-1$ absolute differences involving p_1 increase, each by d. All other terms stay fixed. The net increase is $rd-(r-1)d = d$, proving (5). □

Remark 3. This proves an identity for the *minimal* intersection number. Merely inserting a visible crossing into an immersed picture would not suffice. The strict longest-block hypothesis is important; the lemma does not claim that arbitrary peripheral-letter insertion increases the minimum by one.

4 Many marked classes at one word length

Lemma 4. *For every sufficiently large integer N, there are at least*

$$c_0 \frac{3^N}{N}, \qquad c_0 = \frac{1}{3456},$$

distinct oriented conjugacy classes of length N of the form $a^M u$, satisfying Lemma 2, all with the same marker length $M = M_N$.

Proof. Put

$$R = \lceil \log_3(16N) \rceil, \qquad M = R+1, \qquad t = N - M.$$

Count reduced words u of length t whose first and last letters are b, and which contain neither a^M nor A^M. Let Q be the four-by-four transition matrix with $Q_{xy} = 1$ unless $y = x^{-1}$. Its eigenvalues are $3, 1, 1, -1$, and

$$(Q^{t-1})_{bb} = \frac{3^{t-1} + 2 + (-1)^{t-1}}{4} \geq \frac{3^{t-1}}{4}.$$

Among the 3^{t-1} reduced words starting with b, a run of M specified equal letters at any specified position has probability at most $3^{-(M-1)}$: after its first letter, $M-1$ transitions are forced. A union

bound over at most t starting positions and the two letters a, A bounds the number of bad words by

$$2t\,3^{t-1}3^{-R} \le \frac{1}{8}\,3^{t-1}.$$

There remain at least $3^{t-1}/8 = 3^{N-M}/24$ admissible tails. Since $3^M \le 144N$, this is at least $3^N/(3456N)$.

In $a^M u$, the marked block is the unique longest a-type block. It anchors the cyclic word uniquely: two words constructed in this way are cyclically conjugate only if their tails are equal. Thus the counted words are distinct conjugacy classes, not merely different choices of basepoint. They are primitive by Lemma 2 and nonperipheral: they involve both generators, and their marker of length at least two rules out powers of the third peripheral word. □

Lemma 5. $|\mathcal{F}_N| \sim 3^N/N$ *as* $N \to \infty$.

Proof. The number of cyclically reduced linear words of length N is $\mathrm{tr}(Q^N) = 3^N + 2 + (-1)^N$. The number which are proper powers is $O(N3^{N/2})$, by summing over possible shorter periods. Every remaining cyclic orbit has exactly N elements. Periodic classes contribute an exponentially smaller error, proving the assertion. □

5 Proof of the exact-K lower bound on P

The normal distribution in (4) has variance $1/81$. Its probability of exceeding 1 equals the standard normal probability of exceeding 9, which is less than $c_0/4$. By (4) and Lemma 5, for all sufficiently large N the number of classes satisfying

$$\mathrm{si}(w) > N^2/9 + N^{3/2}$$

is at most $(c_0/2)3^N/N$. Consequently at least

$$\frac{c_0}{2}\frac{3^N}{N} \tag{6}$$

of the marked classes from Lemma 4 satisfy $\mathrm{si}(w) \le N^2/9 + N^{3/2}$.

Given K, choose

$$N = \lfloor 3\sqrt{K} - 9K^{1/4} \rfloor. \tag{7}$$

For all sufficiently large K, $N^2/9 + N^{3/2} \le K$. To check this, put $x = K^{1/4}$. For $x \ge 6 + 3\sqrt{3}$,

$$\frac{(3x^2 - 9x)^2}{9} + (3x^2)^{3/2} = x^4 - (6 - 3\sqrt{3})x^3 + 9x^2 \le x^4.$$

For each good seed $w = a^M u$, set

$$\Phi_K(w) = a^{M+K-\mathrm{si}(w)}u. \tag{8}$$

Lemma 2 gives $\mathrm{si}(\Phi_K(w)) = K$.

This map is injective on the seed conjugacy classes. The unique longest a-type block remains recognizable in every output. Equality of output cyclic words aligns that block and forces equality of the entire tail. The seed marker length M_N is common to the family, so equal tails force equal seeds. In particular, no uncontrolled multiplicity results from the different amounts of padding. All outputs are primitive.

For the thrice-punctured sphere, the orientation-preserving mapping class group is the permutation group of the three punctures; a mapping class fixing all three punctures is isotopic to the identity. Allowing surface orientation reversal gives at most 12 effective transformations on oriented free homotopy classes. Allowing reversal of the curve gives a conservative bound of 24 classes per orbit. For a pair of pants with boundary, the additional boundary twists act trivially on closed free homotopy classes. Thus (6) yields

$$\widetilde{\mathcal{O}}(K,P) \geq \frac{c_0}{48}\frac{3^N}{N}.$$

As $N \geq 3\sqrt{K} - 9K^{1/4} - 1$ and $N \leq 3\sqrt{K}$, this gives

$$\widetilde{\mathcal{O}}(K,P) \geq \frac{1}{1492992\sqrt{K}}\, 3^{\,3\sqrt{K}-9K^{1/4}}$$

for all sufficiently large K. The asymptotic threshold is not numerically determined here, since the proof uses convergence in the central limit theorem.

6 Transfer to the punctured torus and punctured spheres

Lemma 6. *A nontrivial, nonperipheral closed curve in a pair of pants has positive geometric intersection with every essential properly embedded arc.*

Proof. Cutting a pair of pants along an essential properly embedded arc produces one annulus or two annuli. A closed curve disjoint from the arc would lie in one of these annuli and hence be peripheral in the original pair of pants. This is a contradiction. □

Once-punctured torus. Choose an essential simple nonperipheral curve α in $S_{1,1}$, and cut along it. The cut surface P is a pair of pants. Realize α geodesically. A nonperipheral curve in P has its geodesic in the interior of P, so its self-intersection number is unchanged after gluing P back to the torus. Primitivity is also preserved: if the same geodesic were a proper power in the torus, its primitive traversal would still be contained and closed in the interior of P.

For any constructed curve γ, α is the unique isotopy class of essential simple nonperipheral curve disjoint from γ. Indeed, a different such simple curve intersects α and gives an essential proper arc in the cut pants, contradicting Lemma 6. This argument can be performed with geodesic representatives, which simultaneously realize the relevant intersection numbers.

Therefore an ambient mapping class taking one constructed curve to another must preserve $[\alpha]$. After isotopy it restricts to a homeomorphism of the cut pants. Its action on these free homotopy classes consequently has the same finite bound as a pants homeomorphism. The factor 24 above remains valid, proving (1) for $S_{1,1}$.

Important basis distinction: the letters a, b used in this proof are peripheral generators of the *cut pair of pants*. They are not the customary intersecting generator pair of the torus group. No application of the pants intersection formula to a torus basis is being made.

m-punctured spheres, $m \geq 4$. Choose an essential simple curve α cutting off exactly two punctures, and let P be the resulting twice-punctured disk, viewed as a pair of pants. Self-intersection and primitivity are preserved by the same geodesic argument.

The class $[\alpha]$ is characterized by γ as the boundary of a twice-punctured disk carrying this nonperipheral pants curve. To see uniqueness, another such boundary disjoint from γ cannot intersect α, by Lemma 6. If it is disjoint from α and carries γ on its twice-punctured side, it encloses

the same pants side with no extra punctures and is therefore isotopic to α. Equivalently, one can use the unique pants support filled by γ. An ambient homeomorphism carrying one family member to another must preserve this support. For $m = 4$, swapping the two sides would not carry a nonperipheral curve on one side to a class carried on the other: their geodesic representatives have disjoint interiors and a common class could only be peripheral. Thus again there are at most 24 oriented family classes per ambient orbit. This proves the lower bound for all the stated spheres.

7 Upper bound and powers

Theorem 1.1 and Section 7.1 of Aougab–Souto [1] supply the upper bound in (2) for primitive essential curve types on every finite-type orientable surface with negative Euler characteristic; the paper explicitly states that the upper bound extends to punctures. This extension is stronger than the closed-surface-only quotation on page 5 of the supplied notes.

For clarity about conventions, the supplied torus list includes powers such as a^5. For a primitive class δ on an orientable surface the minimal-general-position convention satisfies

$$\mathrm{si}(\delta^q) = q^2 \,\mathrm{si}(\delta) + q - 1$$

(Despré–Lazarus [4, Proposition 26]). Consequently a class with at most K intersections has power exponent at most $K + 1$, and its primitive root has at most K intersections. Allowing powers therefore costs at most a factor $K + 1$, together with the polynomial contribution from powers of simple and peripheral curves. These factors, and the finite orientation/labeling factors, are absorbed in the arbitrary $\varepsilon\sqrt{K}$ slack of (2). This completes the proof of Theorem 1.

8 Algorithmic consequence and relation to the supplied notes

The constructive lower-bound algorithm is: enumerate the marked words at the length (7); compute their minimal self-intersection number; retain those with value at most K; apply (8); and quotient by the finite effective pants action. This produces a large certified subfamily, not all orbit types.

There is also an exhaustive, but not output-optimal, algorithm for $S_{0,3}$. The word-length lower bound of Chas–Phillips [5] implies that a class of self-intersection at most K has cyclic length at most $2K + 2$ in the pants basis (peripheral powers may alternatively be handled directly). Enumerate all such cyclic words, compute their intersection numbers, filter for exactly K, and quotient by the finite pants action. This is exponential in K as a naive enumeration, and should not be confused with a sharp running-time bound.

The useful ingredients of the supplied documents are the separation of minimality from orbit identification, the passage to free-group words, and the warning about triangle moves. Raw ribbon/basepoint graphs do not automatically classify free homotopy orbits. In particular the supplied torus list is a candidate list, not a verified count of 44 distinct orbits: the entries `abaaBB` and `BabaaB` are cyclic rotations of the same word. The proposed $k = 4$ basepoint normal-form proof is explicitly unfinished in that file. Neither that normal form nor the number 44 is assumed here.

A precise asymptotic equivalent, an optimal leading exponential coefficient, and a full orbit-enumeration algorithm running in $\exp(O(\sqrt{K}))$ time are not established in this note. The exact-K padding/injectivity argument is the substantive additional step beyond using a cumulative count.

A Definition of the block-boundary term H

This appendix records the published input (3) in a checkable form. Write $w = s_1^{p_1} t_1^{q_1} \cdots s_r^{p_r} t_r^{q_r}$ as above. For each i, let x_i be the cyclic rotation beginning at the first letter of $s_i^{p_i}$, and let y_i be the rotation beginning at the first letter of $t_i^{q_i}$. In the planar Schottky coding, let α_i have endpoints $x_i^{+\infty}, x_i^{-\infty}$ and let β_i have endpoints $y_i^{+\infty}, y_i^{-\infty}$; the latter notation means infinite repetition of the inverse word when the exponent is negative. These are lifts based at block boundaries.

Set

$$
\begin{aligned}
C_i^1 &= \{j : i < j \le r,\ \alpha_i \cap \alpha_j \neq \varnothing,\ s_i^{p_i} \neq s_j^{p_j}\},\\
C_i^2 &= \{j : i < j \le r,\ \beta_i \cap \alpha_j \neq \varnothing,\ s_i^{p_i} \neq (s_j^{-1})^{p_j}\},\\
D_i^1 &= \{j : i < j \le r,\ \beta_i \cap \beta_j \neq \varnothing,\ t_i^{q_i} \neq t_j^{q_j}\},\\
D_1^2 &= \{j : 1 \le j \le r,\ \alpha_1 \cap \beta_j \neq \varnothing\},\\
D_i^2 &= \{j : i \le j \le r,\ \alpha_i \cap \beta_j \neq \varnothing,\ t_{i-1}^{q_{i-1}} \neq (t_j^{-1})^{q_j}\} \quad (i \ge 2),
\end{aligned}
$$

and

$$
H(w) = \sum_{i=1}^{r} \big(|C_i^1| + |C_i^2| + |D_i^1| + |D_i^2|\big).
$$

Two geodesics cross exactly when their four ideal endpoints alternate on the circle. The first-divergence endpoint ordering used in Lemma 2 is the Birman–Series ordering quoted in Section 3.1 of [3]. The formula is used here only for nonpowers, avoiding conventions for multiply traversed geodesics.

References

[1] T. Aougab and J. Souto, *Counting curve types*, arXiv:1606.06067. In particular Theorem 1.1 and Section 7.1.

[2] M. Chas and S. P. Lalley, *Self-intersections in combinatorial topology: statistical structure*, Inventiones Mathematicae **188** (2012), 429–463; arXiv:1012.0580. Main Theorem.

[3] E. H. A. A. Diop and M. Gaye, *Self-intersection on Pair of Pants*, in *Nonlinear Analysis, Geometry and Applications*, Springer (2022), 429–454; preprint *Self intersections on a pair of pants*, arXiv:2108.06796. Section 3, especially Theorem 3.12 in the preprint.

[4] V. Despré and F. Lazarus, *Computing the geometric intersection number of curves*, Journal of the ACM **66**(6) (2019), Article 45; arXiv:1511.09327. Proposition 26.

[5] M. Chas and A. Phillips, *Self-intersection numbers of curves in the doubly-punctured plane*, Experimental Mathematics **21**(1) (2012), 26–37; arXiv:1001.4568.

[6] M. Genkin and J. Suk, *Computing Mapping Class Group Orbits of Closed Curves on Hyperbolic Surfaces*, June 9, 2017, supplied research notes; and the supplied manuscript *Punctured Torus Count.tex*.